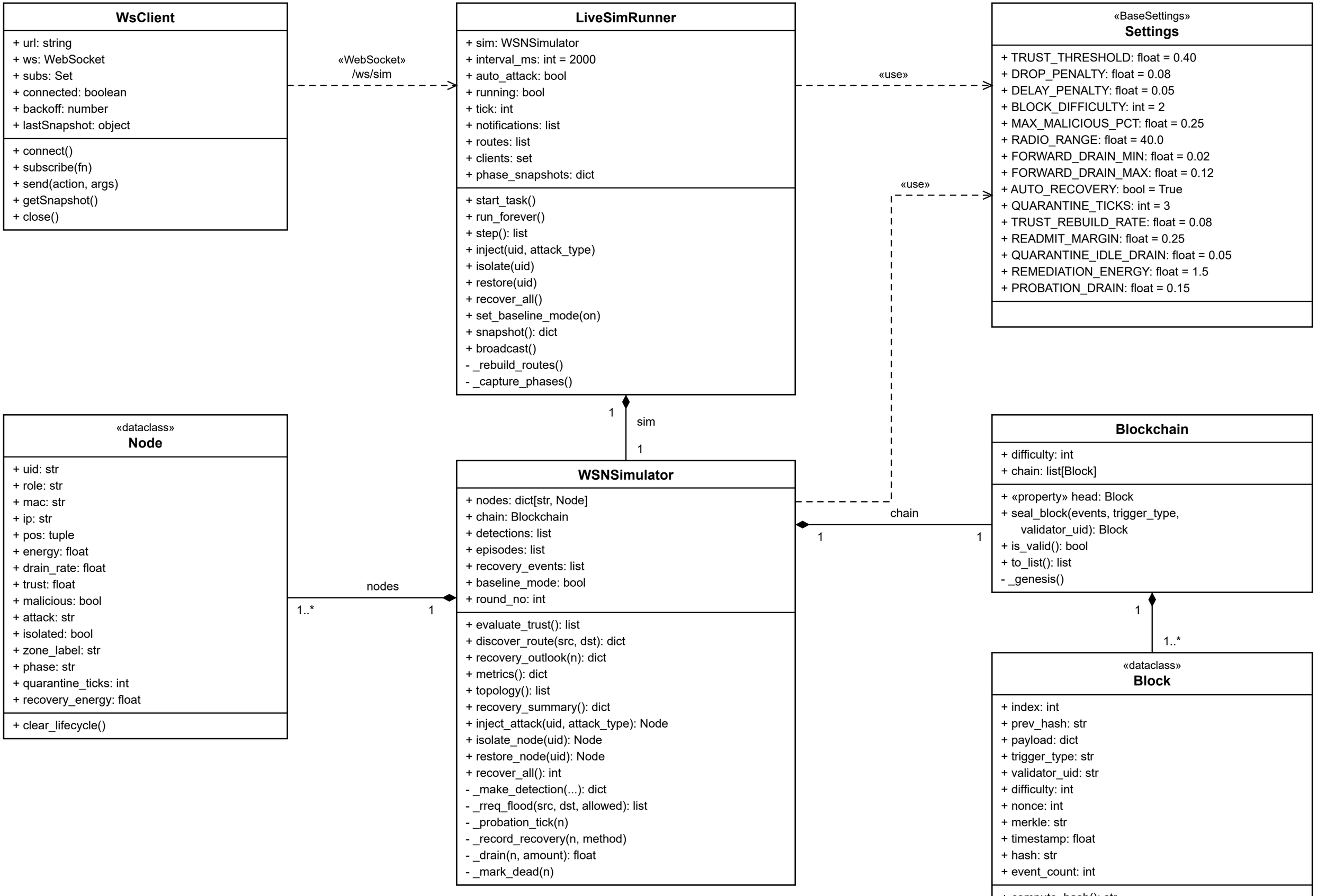

WsClient
+ url: string
+ ws: WebSocket
+ subs: Set
+ connected: boolean
+ backoff: number
+ lastSnapshot: object
+ connect()
+ subscribe(fn)
+ send(action, args)
+ getSnapshot()
+ close()
«WebSocket»
/ws/sim
LiveSimRunner
+ sim: WSNSimulator
+ interval_ms: int = 2000
+ auto_attack: bool
+ running: bool
+ tick: int
+ notifications: list
+ routes: list
+ clients: set
+ phase_snapshots: dict
+ start_task()
+ run_forever()
+ step(): list
+ inject(uid, attack_type)
+ isolate(uid)
+ restore(uid)
+ recover_all()
+ set_baseline_mode(on)
+ snapshot(): dict
+ broadcast()
- _rebuild_routes()
- _capture_phases()
«use»
«BaseSettings»
Settings
+ TRUST_THRESHOLD: float = 0.40
+ DROP_PENALTY: float = 0.08
+ DELAY_PENALTY: float = 0.05
+ BLOCK_DIFFICULTY: int = 2
+ MAX_MALICIOUS_PCT: float = 0.25
+ RADIO_RANGE: float = 40.0
+ FORWARD_DRAIN_MIN: float = 0.02
+ FORWARD_DRAIN_MAX: float = 0.12
+ AUTO_RECOVERY: bool = True
+ QUARANTINE_TICKS: int = 3
+ TRUST_REBUILD_RATE: float = 0.08
+ READMIT_MARGIN: float = 0.25
+ QUARANTINE_IDLE_DRAIN: float = 0.05
+ REMEDIATION_ENERGY: float = 1.5
+ PROBATION_DRAIN: float = 0.15
1
sim
1
«use»
«dataclass»
Node
+ uid: str
+ role: str
+ mac: str
+ ip: str
+ pos: tuple
+ energy: float
+ drain_rate: float
+ trust: float
+ malicious: bool
+ attack: str
+ isolated: bool
+ zone_label: str
+ phase: str
+ quarantine_ticks: int
+ recovery_energy: float
+ clear_lifecycle()
nodes
1..*
1
WSNSimulator
+ nodes: dict[str, Node]
+ chain: Blockchain
+ detections: list
+ episodes: list
+ recovery_events: list
+ baseline_mode: bool
+ round_no: int
+ evaluate_trust(): list
+ discover_route(src, dst): dict
+ recovery_outlook(n): dict
+ metrics(): dict
+ topology(): list
+ recovery_summary(): dict
+ inject_attack(uid, attack_type): Node
+ isolate_node(uid): Node
+ restore_node(uid): Node
+ recover_all(): int
- _make_detection(...): dict
- _rreq_flood(src, dst, allowed): list
- _probation_tick(n)
- _record_recovery(n, method)
- _drain(n, amount): float
- _mark_dead(n)
chain
1
1
Blockchain
+ difficulty: int
+ chain: list[Block]
+ «property» head: Block
+ seal_block(events, trigger_type,
validator_uid): Block
+ is_valid(): bool
+ to_list(): list
- _genesis()
1
1..*
«dataclass»
Block
+ index: int
+ prev_hash: str
+ payload: dict
+ trigger_type: str
+ validator_uid: str
+ difficulty: int
+ nonce: int
+ merkle: str
+ timestamp: float
+ hash: str
+ event_count: int
+ compute_hash(): str
+ mine(): Block